Exact-Rational Certificates in Podium: Constructions, Proofs, and Prior Art

A technical note

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Abstract

This note collects the constructions, proofs, and prior-art positioning behind the exact-rational certificates in the Podium library (`podium.verify`): barrier certificates for abort safety, Karush–Kuhn–Tucker certificates for the online convex solves, control-Lyapunov certificates, sum-of-squares certificates, and an optimality-gap bracket for nonconvex quadratically constrained quadratic programs. None of the underlying mathematics is new; the note documents the exact instantiation, its verification, and its relation to the exact-SOS/moment, rational-SDP-recovery, and QCQP-duality literature, so that a math-inclined reader can audit the certificate layer directly. Each result is exercised by a test in the repository. In detail, guidance and control software relies on numerical optimization and floating-point synthesis whose executions are difficult to verify directly. We study the problem of verifying solver *outputs*: given a certificate produced by an untrusted floating-point solver, we re-verify it in exact rational arithmetic, so that acceptance constitutes a proof that carries no rounding error. We apply this method to four properties: abort safety, via barrier certificates; optimality of the online convex solves, via Karush–Kuhn–Tucker (KKT) conditions for quadratic and second-order cone programs; closed-loop stability, via Lyapunov ellipsoids from a discrete-time linear-quadratic regulator; and polynomial set containment, via sum-of-squares certificates. These certificates are standard; we give exact re-verification procedures — tolerance-free for the barrier, Lyapunov, sum-of-squares, and optimality-gap certificates, and an exact suboptimality bound for the KKT re-check of an approximate online solve — and a reference implementation.

For nonconvex quadratically constrained quadratic programs (QCQPs), we specialize the exact rational recovery of a semidefinite certificate, developed for polynomial optimization by Peyrl and Parrilo and by Kaltofen et al., to the S-procedure dual. This yields a tolerance-free, machine-checkable optimality-gap bracket: a rational lower bound from the S-procedure dual and a rational upper bound from a feasible point, coinciding exactly at a global optimum. Our analytic contribution is a characterization of how the certified lower bound degrades under rational rounding of the dual multiplier: the value loss is second-order in the rounding denominator in the nonsingular case and first-order in the trust-region hard case, with soundness retained in both, and it vanishes when the optimal multiplier is rational (Theorems 2 and 3). For several constraints the S-procedure need not be tight, and the bracket then encloses $J^\star$ between exact rational bounds, certifying a strictly positive suboptimality gap on a Celis–Dennis–Tapia instance (Theorem 4). Each result is accompanied by a verifying test in the reference implementation.

*Starcloud, Inc. Companion to the repository `github.com/adi-oltean/podium`. This technical note documents the exact-arithmetic certificates in `podium.verify` and their proofs; the underlying certificate mathematics is standard, with a modest original analysis of the rational-rounding degradation (see the positioning at the end of Section 2). Cite as `https://doi.org/10.5281/zenodo.21247380`.

1 Introduction

Software increasingly makes consequential decisions by solving optimization problems numerically. A spacecraft plans a fuel-optimal approach; a controller selects a maneuver that must avoid a keep-out zone; an autopilot commits to a command computed by a convex solver. In each case the solver returns an answer, but the answer is a floating-point number produced by an opaque computation rather than a proof that the answer is correct or optimal. Floating-point arithmetic rounds at every step, and a computation that appears to succeed can return a value that is subtly wrong. Where a mistake is costly and cannot be undone after the fact, as in spaceflight, it is desirable for the solver's answer to carry a *certificate*: a small, separate object that an independent program can verify exactly, so that correctness rests on the verifier rather than on the solver. This paper constructs such certificates for guidance and control, and studies in particular the problem of certifying that a computed trajectory is close to the best one achievable.

A concrete instance motivates the main results. Nonconvex trajectory optimization for rendezvous, atmospheric entry, and powered landing is commonly solved by successive convexification, which linearizes the nonconvex constraints about the current iterate and solves a convex subproblem, repeating until convergence [1]. The method returns a feasible trajectory, but that trajectory is only locally optimal, and the solver reports its cost as a floating-point number rather than a bound on the global optimum. The method thus provides no bound on the distance between the computed cost and the global optimum. This paper supplies such a bound: an exact-arithmetic certificate that brackets the global optimum, so that the suboptimality of a computed solution is certified rather than assumed, and, when the bracket closes, the solution is certified globally optimal. The certificate requires neither convexity nor trust in the solver.

The underlying idea is general. A guidance and control stack executes a large amount of untrusted numerical code: convex solvers compute commands, semidefinite programs (SDPs) synthesize safety and stability certificates, and floating-point routines emit actuator signals. Verifying the *execution* of these components, including worst-case iteration counts and bitwise solver behavior, is difficult and often impractical [2]. We instead verify the *answer*: an untrusted floating-point tool proposes a certificate, and a trusted checker re-establishes the property in exact rational arithmetic, without floating-point computation of its own. Because the check is a short, deterministic computation, it can be re-run automatically whenever the code or data change. This matters in practice: a candidate certificate produced in floating point may satisfy its defining condition only approximately, so a tolerance-based check can accept a pair (λ, t) whose value t exceeds the true optimum, whereas the exact rational test accepts a certificate only when it holds exactly.

Contributions.

1. An exact-rational verification layer for four standard guidance and control certificates, barrier abort safety, Karush–Kuhn–Tucker (KKT) optimality of the online convex solves, Lyapunov stability, and sum-of-squares set containment (Section 3), each synthesized by an untrusted solver and re-checked over the rationals $\mathbb{Q}$ (exactly, or within an explicit rational bound for the approximate online solves).

2. A specialization of exact rational SDP-certificate recovery to the S-procedure dual of a nonconvex quadratically constrained quadratic program (QCQP), giving a tolerance-free machine-checkable optimality-gap bracket for guidance and control (Section 4). This furnishes the certified global-suboptimality bound that a successive convexification solution does not carry (Section 4.5).

3. A rounding-rate analysis of that bracket: the certified lower bound loses value at second order in the rounding denominator in the nonsingular case (Theorem 2) and at first order in the trust-region hard case (Theorem 3), with soundness retained in both, and the bracket certifies an exact duality gap when the S-procedure is not tight (Theorem 4).
4. An open reference implementation whose trusted checker uses no floating point, in which each result is exercised by a test that reconstructs the certificate over $\mathbb{Q}$ (Section 5).

None of the underlying certificates is new. The barrier certificates are due to Prajna and Jadbabaie [3] and Ames et al. [4]; the Lyapunov re-verification follows Wang et al. [5]; the S-lemma is due to Yakubovich [6] and its exactness is surveyed by Pólik and Terlaky [7]. Most importantly, the recovery of an exact rational certificate from a floating-point SDP solution, and its use to certify global lower bounds in polynomial optimization, is due to Peyrl and Parrilo [8], Kaltofen et al. [9], and Magron and Safey El Din [10]; Davis and Papp [11] recover exact rational certificates directly from a numerical dual certificate and prove a convergence rate, and rigorous bracketing of a relaxation value is provided by verified SDP [12, 13]. A nonconvex QCQP is the degree-two case of this machinery, and its exactness and hidden convexity are themselves classical [14, 15, 7]. Our contribution is therefore not exact rational certification, the bracket idea, nor certificate recovery with a rate, but the specialization of these to the S-procedure dual of a QCQP, in particular the second-order (nonsingular) versus first-order (trust-region hard case) rounding-rate dichotomy of Section 4, together with a tolerance-free implementation for a guidance and control verification stack.

2 Exact re-verification of solver output

A *certificate* is a small object that makes a property easy to confirm even when the property was hard to establish: a Lyapunov matrix for stability, a multiplier for a bound, a sum-of-squares decomposition for nonnegativity. Our method separates search from proof. An untrusted floating-point routine searches for the certificate, typically by solving an SDP or an interior-point problem. The result is then rounded to nearby rational numbers and checked by an exact computation over $\mathbb{Q}$.

The single trusted primitive is an exact test for positive semidefiniteness (PSD), a symmetric $LDL^\top$ elimination over $\mathbb{Q}$ in $O(n^3)$ operations: a symmetric matrix is PSD if and only if every pivot is nonnegative and each zero pivot leaves the rest of its row, and hence by symmetry its column, in the Schur complement zero. The resulting factorization $M = LDL^\top$ with $D \succeq 0$ is itself the nonnegativity witness, since $z^\top M z = \sum_i D_{ii}(l_i^\top z)^2$. This avoids eigenvalues, which are in general irrational and admit no exact rational representation, and the leading-minor test of Sylvester, which certifies strict definiteness only. The trusted checker uses only exact rational arithmetic, so its conclusions carry no rounding error. For the barrier, Lyapunov, sum-of-squares, and optimality-gap certificates, acceptance is an exact identity together with the PSD test, with no tolerance. When the object under test is the output of an approximate solver, as for the online convex solves below, the residuals are still computed exactly, but a rigorous bound on suboptimality is reported only when those exact residuals yield a valid Lagrangian dual point, not merely when they are small.

3 Four certificate families

Barrier certificates for abort safety. For linearized relative motion with a spherical keep-out zone, an SDP produces a barrier function whose sublevel set is invariant under the passive, thrust-free drift. The checker verifies over $\mathbb{Q}$ an S-procedure condition, in which a nonnegative multiplier certifies that one quadratic is nonnegative on the region where another is nonnegative. The barrier is expressed on a basis of quantities conserved by the passive drift, so its Lie derivative — the rate of change of the barrier along the drift — vanishes by construction. Together these establish forward invariance of the safe set, so that the abort set never intersects the keep-out zone [3, 4].

KKT certificates for the online solves. The guidance layer solves a quadratic program (QP) and a second-order cone program (SOCP) online. The solver returns an approximate primal-dual point, and the checker computes its KKT residuals, stationarity, primal and dual feasibility, and complementary slackness, exactly. A small stationarity residual is not by itself a bound on suboptimality: with a rank-deficient objective (a linear or minimum-fuel program) an approximately stationary point can be arbitrarily far from optimal. The checker therefore reports a rigorous bound only when a valid Lagrangian dual point exists, namely when the multiplier is dual-feasible, the Hessian P is PSD, and the stationarity residual r lies in the range of P. The exact gap $\mu^\top(h - Gx) + \nu^\top(b - Ax) + \frac{1}{2}r^\top P^+ r$ then bounds $p(x) - p^\star$, the term $\frac{1}{2}r^\top P^+ r$ being the objective drop from the candidate to the exact minimizer of the Lagrangian, computed by solving $Pz = r$ over the rationals. A SOCP objective is linear and so has no curvature to absorb a residual; a rigorous bound there instead requires exact conic-dual feasibility. Membership of a slack vector $s = (s_0, s_{1:}) \in \mathbb{R} \times \mathbb{R}^d$ in the second-order cone is decided without square roots through the algebraic condition $s_0 \geq 0$ and $\|s_{1:}\|^2 \leq s_0^2$. In every case the checker verifies that the objective Hessian is PSD, so that the certificate concerns a global optimum of a convex program rather than a stationary point of unknown type.

Lyapunov certificates for closed-loop stability. A cost-to-go matrix P from a discrete-time linear-quadratic regulator (LQR) is re-verified as positive definite, so that its sublevel sets are bounded ellipsoids, and as satisfying the Lyapunov decrease condition $P - A_{\mathrm{cl}}^\top P A_{\mathrm{cl}} \succeq 0$, where $A_{\mathrm{cl}} = A - BK$ is the closed-loop transition matrix under the state-feedback gain K, using the same exact PSD test, in the sense of [5].

Sum-of-squares (SOS) certificates for set containment. The nonnegativity of a polynomial $p(x)$, for instance that a certified attitude cone contains a given sublevel set, is verified as an identity $p(x) = z(x)^\top G\, z(x)$ with G symmetric positive semidefinite, where $z(x)$ is a vector of monomials and G is the associated Gram matrix. Such a representation exhibits p as a sum of squares, and hence as nonnegative. The Gram matrix is recovered from an approximate SDP solution by the rational rounding of Peyrl and Parrilo [8] and Roux et al. [13].

4 Optimality-gap bounds for nonconvex QCQPs

Quadratically constrained quadratic programs arise throughout guidance: keep-out zones, thrust limits, and terminal sets are quadratic, and each convex subproblem inside a successive convexification loop is a convex QCQP or its second-order cone relaxation. When the constraints are nonconvex the problem is NP-hard in general, and a solver can return a feasible point but not, on its own, a proof that no better point exists. We construct such a proof in exact arithmetic: a lower bound t and an upper bound u with $t \leq J^\star \leq u$, both rational and both checked without floating point, so that $u - t$ is a certified optimality gap and the case $t = u$ certifies a global optimum (Figure 1).

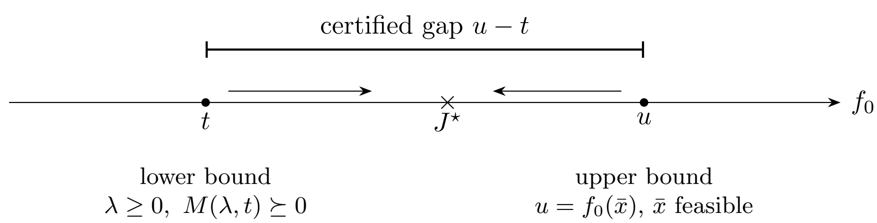

Figure 1: The bracket. A rational multiplier with $M(\lambda, t) \succeq 0$ gives a certified lower bound $t \leq J^\star$; an exactly feasible point gives a certified upper bound $u = f_0(\bar{x}) \geq J^\star$. Both are exact rationals, so $u - t$ is a certified optimality gap; when $t = u$ the global optimum is certified exactly. The arrows indicate refinement of the dual and of the feasible point.

Consider the QCQP

$$J^\star = \inf_{x \in \mathbb{R}^n} \Big\{ f_0(x) : f_k(x) \geq 0,\ k = 1, \ldots, m \Big\}, \qquad f_j(x) = x^\top P_j x + q_j^\top x + r_j, \tag{1}$$

with symmetric P_j. A keep-out constraint $\|x - c\| \geq R$ renders the feasible set nonconvex. Throughout we use the instance $\min \|x\|^2$ subject to $\|x - (3, 0)\| \geq 5$, whose origin lies inside the excluded ball; its global optimum is $J^\star = 4$, attained at $(-2, 0)$ with multiplier $\lambda^\star = 2/5$. Table 1 lists this and the other cases considered below.

4.1 The bracket and its soundness

The lower bound is a Lagrangian relaxation, certified exactly. For multipliers $\lambda \in \mathbb{R}^m_{\geq 0}$ the Lagrangian $L(x, \lambda) = f_0(x) - \sum_k \lambda_k f_k(x)$ is quadratic in x, and with the homogenized coordinate $z = [x; 1] \in \mathbb{R}^{n+1}$ the shifted form $L(x, \lambda) - t$ equals $z^\top M(\lambda, t) z$, where

$$M(\lambda, t) = \begin{bmatrix} P_0 - \sum_k \lambda_k P_k & \frac{1}{2}(q_0 - \sum_k \lambda_k q_k) \\ \frac{1}{2}(q_0 - \sum_k \lambda_k q_k)^\top & r_0 - \sum_k \lambda_k r_k - t \end{bmatrix}. \tag{2}$$

Theorem 1 (Lower bound)**.** *If $\lambda \geq 0$ componentwise and $M(\lambda, t) \succeq 0$, then $t \leq J^\star$.*

Proof. For any x, writing $z = [x; 1]$, expansion gives $z^\top M(\lambda, t) z = f_0(x) - \sum_k \lambda_k f_k(x) - t$. Positive semidefiniteness of M makes this nonnegative for all x. At a feasible x, each $f_k(x) \geq 0$ and each $\lambda_k \geq 0$, so $\sum_k \lambda_k f_k(x) \geq 0$ and hence $f_0(x) \geq t$. Taking the infimum over feasible x gives $t \leq J^\star$. $\square$

The argument is Lagrangian weak duality and uses no convexity, constraint qualification, or boundedness. Conversely, on the primal side, any point $\bar{x}$ with $f_k(\bar{x}) \geq 0$ for all k gives $J^\star \leq f_0(\bar{x})$. The feasibility of $\bar{x}$ must be checked exactly: a point that is feasible only within a solver tolerance may have objective value below $J^\star$, which would place the upper bound below the true optimum. The checker therefore evaluates $f_k(\bar{x}) \geq 0$ over the rationals. By elementary gap closure, if a certified lower bound t equals a certified upper bound $u = f_0(\bar{x})$ then $J^\star = t = u$, an exact certificate of the global optimum. The two bounds are the rationalized Shor/S-procedure lower bound and a rational feasible point, standard ingredients of verified global optimization; the content of this section is their specialization to the QCQP dual and the rounding-rate analysis below.

4.2 Recovery of an exact bound from an approximate dual

Fix $m = 1$ and write $A(\lambda) = P_0 - \lambda P_1$, $b = q_0 - \lambda q_1$, $c = r_0 - \lambda r_1$, and $g(\lambda) = \inf_x (f_0 - \lambda f_1)$ on the set $U = \{\lambda : A(\lambda) \succ 0\}$.

Lemma 1. *For $\lambda \in U$, $g(\lambda) = c - \frac{1}{4}b^\top A(\lambda)^{-1}b$, and $M(\lambda, t) \succeq 0$ if and only if $t \leq g(\lambda)$.*

Proof. On U the quadratic $f_0 - \lambda f_1 = x^\top Ax + b^\top x + c$ is strictly convex, with unique minimizer $x^\star = -\frac{1}{2}A^{-1}b$ and minimum value $g(\lambda) = c - \frac{1}{4}b^\top A^{-1}b$. Since $A \succ 0$, the Schur complement[1] of A in $M(\lambda, t)$ is $(c - t) - \frac{1}{4}b^\top A^{-1}b$, and $M(\lambda, t) \succeq 0$ if and only if this is nonnegative, that is, $t \leq g(\lambda)$. □

By Cramer's rule g is a rational function of λ with coefficients rational in the problem data, so a rational $\lambda \in U$ yields a rational, and therefore exactly certified, lower bound. For a single constraint with a strictly feasible point, the S-lemma gives strong duality, $J^\star = \max_{\lambda \geq 0} g(\lambda) = g(\lambda^\star)$ [6].

Theorem 2 (Nonsingular recovery)**.** *Suppose $A(\lambda^\star) \succ 0$ and $\lambda^\star > 0$ is an interior maximizer of the dual. Then g is smooth and concave near $\lambda^\star$ with $g'(\lambda^\star) = 0$, so $J^\star - g(\lambda) = O((\lambda - \lambda^\star)^2)$. Rounding $\lambda^\star$ to a rational λ with denominator at most D, so that $|\lambda - \lambda^\star| \leq 1/D$, yields an exactly certified lower bound with $J^\star - g(\lambda) = O(1/D^2)$. If in addition $\lambda^\star$ and $J^\star$ are rational, then $(\lambda^\star, J^\star)$ is an exact certificate, and with a rational optimizer the bracket closes exactly.*

Proof. On U, g is a rational function, hence smooth, and is concave as an infimum of functions affine in λ. At an interior maximizer $g'(\lambda^\star) = 0$, so Taylor's theorem gives $J^\star - g(\lambda) = \frac{1}{2}|g''(\xi)|(\lambda - \lambda^\star)^2$ for some ξ between λ and $\lambda^\star$. Substituting $|\lambda - \lambda^\star| \leq 1/D$ gives the $O(1/D^2)$ bound. Exactness of $g(\lambda^\star)$ when $\lambda^\star$ is rational follows from Lemma 1. □

Remark. With several constraints, an interior maximizer means interior with respect to the active constraints. The stationarity $\partial g/\partial \lambda_k = 0$ holds for active multipliers ($\lambda_k^\star > 0$); an inactive constraint sits at $\lambda_k^\star = 0$ on the boundary of the nonnegative orthant, where the corresponding derivative is negative.

In the reference implementation, rounding an approximate dual value $\tilde{\lambda} \approx 0.4003$ for the running instance to a denominator at most 100 recovers $\lambda^\star = 2/5$ and $t = 4 = J^\star$ exactly.

4.3 The singular case

Theorem 3 (Singular recovery)**.** *Suppose $A(\lambda^\star)$ is positive semidefinite but singular, the trust-region hard case [16]. Then: (a) if $\lambda^\star$ and $J^\star$ are rational, $(\lambda^\star, J^\star)$ is an exact certificate, since M is a rank-deficient PSD matrix that the exact test accepts, so soundness is unaffected; (b) approaching $\lambda^\star$ from within U gives certified rational lower bounds that converge to $J^\star$ linearly; (c) exact closure requires $\lambda^\star$ and $J^\star$ to be rational, which does not hold generically, as $J^\star$ may be algebraic of degree greater than one.*

Proof. (a) Soundness is Theorem 1, which requires only $\lambda^\star \geq 0$ and $M(\lambda^\star, J^\star) \succeq 0$; the exact $LDL^\top$ test accepts a rank-deficient PSD matrix, since at a zero pivot the rest of the row vanishes and the elimination continues with a nonnegative D. (b) Since $A \succ 0$ on U and $A(\lambda^\star)$ is singular, $\lambda^\star$ lies on the boundary of U. On U, g is the smooth concave function of Lemma 1, and $J^\star = \sup_{\lambda \geq 0} g$ is attained at $\lambda^\star$ on that boundary. A boundary maximizer need not be stationary, so the right derivative $g'_+(\lambda^\star) := \lim_{\lambda \downarrow \lambda^\star} g'(\lambda)$ is generically nonzero, and a first-order expansion gives $J^\star - g(\lambda) = |g'_+(\lambda^\star)|\,|\lambda - \lambda^\star| + o(|\lambda - \lambda^\star|)$, which is linear. This contrasts with Theorem 2, where an interior maximizer forces $g'(\lambda^\star) = 0$ and a quadratic rate. (c) Here $\lambda^\star$ solves $\det A(\lambda) = 0$ together with the boundary optimality condition, so it may be algebraic of degree greater than one, and rationality of $\lambda^\star$ and $J^\star$ fails in general. □

[1] For a block matrix $\left[\begin{smallmatrix} A & v \\ v^\top & d \end{smallmatrix}\right]$ with $A \succ 0$, positive semidefiniteness is equivalent to nonnegativity of the scalar Schur complement $d - v^\top A^{-1}v$.

Soundness is thus retained in the singular case; only the recovery rate falls. The implementation's instance $\min -x_1^2 + 2x_2^2 + 2x_2$ subject to $1 - \|x\|^2 \geq 0$ has $A(\lambda) = \operatorname{diag}(-1+\lambda,\, 2+\lambda)$, singular at $\lambda^\star = 1$, with $J^\star = -4/3$. The certified gap decreases by one order of magnitude for each order of magnitude of $|\lambda - 1|$, consistent with the linear rate, compared with two orders in the nonsingular case. An exact rational recovery in the singular case remains open.

4.4 Several constraints

Theorem 4 (Certified duality gap)**.** *Theorem 1 and the closing observation hold for every m. Let $d^\star$ be the Shor relaxation value of (1), that is, the supremum of t over rational $\lambda \geq 0$ and t with $M(\lambda, t) \succeq 0$. Then $d^\star \leq J^\star$ always, and for $m \geq 2$ this inequality may be strict. When it is, any certified lower bound $t \leq d^\star$ and any exactly feasible $\bar{x}$ give an interval $[t, f_0(\bar{x})]$ that contains $J^\star$, with both endpoints exact rationals; $f_0(\bar{x}) - t$ is then a certified upper bound on the suboptimality of $\bar{x}$, and hence on the duality gap $J^\star - d^\star$.*

Proof. The lower-bound argument of Theorem 1 uses only that a multiplier $\lambda \geq 0$ with $M(\lambda, t) \succeq 0$ makes $f_0(x) - \sum_{k=1}^m \lambda_k f_k(x) - t$ a global quadratic minorant; it does not use $m = 1$. For any feasible x one has $f_k(x) \geq 0$ and $\lambda_k \geq 0$, so

$$f_0(x) - t \;\geq\; \sum_{k=1}^{m} \lambda_k \, f_k(x) \;\geq\; 0,$$

hence $t \leq f_0(x)$ for every feasible x, and therefore $t \leq J^\star$. Taking the supremum over rational $\lambda \geq 0$ and t with $M(\lambda, t) \succeq 0$ gives $d^\star \leq J^\star$.

That the inequality can be strict for $m \geq 2$ is classical: exactness of the S-procedure is a two-quadratic phenomenon—the S-lemma holds for a single constraint but not in general for two or more [7]—so the Shor relaxation of a multi-constraint QCQP need not be tight; the Celis–Dennis–Tapia family below furnishes instances with $d^\star < J^\star$.

Finally, let t be a certified lower bound, so $\lambda \geq 0$ and $M(\lambda, t) \succeq 0$ pass the exact PSD test, and let $\bar{x}$ be exactly feasible, so $f_k(\bar{x}) \geq 0$ for every k, each checked over $\mathbb{Q}$. Then $t \leq d^\star \leq J^\star \leq f_0(\bar{x})$, so $J^\star \in [t, f_0(\bar{x})]$ with both endpoints exact rationals, and $f_0(\bar{x}) - t \geq J^\star - d^\star \geq 0$ is a certified upper bound on both the suboptimality of $\bar{x}$ and the duality gap. ☐

Remark. For $m = 1$ with a strictly feasible point the S-lemma gives $d^\star = J^\star$, so the bracket can close (Theorem 2). For $m \geq 2$ this strong duality may fail. Higher levels of the moment hierarchy of Lasserre [17] then give nondecreasing certified lower bounds converging to $J^\star$ under a compactness (Archimedean) condition, each verified by an exact PSD test.

The implementation exhibits the strict case on the two-ball Celis–Dennis–Tapia instance $\min 2x_1x_2$ subject to $\|x - (1,0)\| \leq 2$ and $\|x + (1,0)\| \leq 2$,[2] a trust-region hard case in which the largest certifiable lower bound $d^\star = -3$ lies strictly below the optimum $J^\star \approx -1.476$; the bracket $[-3, -17520/11881]$ then encloses this nonconvex optimum, which has no closed form, with both endpoints verified over $\mathbb{Q}$.

4.5 Relation to successive convexification

Nonconvex trajectory optimization in guidance is commonly addressed by successive convexification and sequential convex programming, which solve a nonconvex program by repeatedly convexifying

[2] The Celis–Dennis–Tapia problem minimizes a quadratic over the intersection of two quadratic constraints; it is the standard example in which the Shor relaxation can be inexact.

Table 1: Representative instances. All certified bounds are exact rationals verified over $\mathbb{Q}$; "closed" indicates that the certified lower bound t and upper bound u coincide and hence certify $J^\star$. The value $\tilde{\lambda} \approx 0.4003$ is the approximate dual fed to the rational rounding step; the recovered certificate is exact.

Instance	Case	$\lambda^\star$	certified $[t,\, u]$	$J^\star$
$\min \lVert x \rVert^2,\ \lVert x-(3,0)\rVert \geq 5$	nonsingular, closed	2/5	[4, 4]	4
same, non-optimal upper point	open bracket	2/5	[4, 169/25]	4
recovery from $\tilde{\lambda} \approx 0.4003$	quadratic rate	2/5	$t = 4$	4
$\min -x_1^2 + 2x_2^2 + 2x_2,\ \lVert x \rVert \leq 1$	singular	1	$t = -4/3$	$-4/3$
two active keep-outs	several constraints	$(\frac{1}{2}, \frac{1}{2})$	[5, 9]	5
Celis–Dennis–Tapia $\min 2x_1x_2$	duality gap	$(\frac{1}{2}, \frac{1}{2})$	$[-3,\, -\frac{17520}{11881}]$	≈ -1.476

about the current iterate and solving a convex subproblem [1, 18]. Such methods return a feasible trajectory that is locally optimal, without a bound on its distance from the global optimum. The bracket supplies that bound. A converged iterate $\bar{x}$ is a feasible point, hence a certified upper bound $u = f_0(\bar{x})$; the S-procedure dual of Section 4 supplies a certified lower bound $t \leq J^\star$ for the same problem (Theorem 1); and $[t, u]$ is a bound on the global suboptimality of the trajectory, with $u - t$ its certified gap, all verified over $\mathbb{Q}$. When the bracket closes, the trajectory is certified globally optimal; when it does not, the residual gap $u - t$ is still a certified bound on suboptimality, and Theorem 4 identifies cases in which the lower bound cannot be tight. We establish this on the keep-out QCQP, the nonconvex primitive that these methods convexify.

5 Reference implementation

The certificates and theorems above are implemented in a reference library whose trusted checker uses only exact rational arithmetic. Each theorem is exercised by a test that constructs the certificate and re-verifies it over $\mathbb{Q}$; the representative cases are collected in Table 1. Two requirements that a tolerance-based check would miss are enforced exactly. First, the checker verifies feasibility of $\bar{x}$ over the rationals, since a point feasible only within a tolerance may have objective value below $J^\star$. Second, the closing criterion draws its two bounds from a single problem instance, since bounds computed for different instances cannot be compared.

6 Related work

Exact and certified certificates for global polynomial optimization are an established subject. The moment/SOS hierarchy of Lasserre [17] and Parrilo [19] gives convergent certified lower bounds, Henrion and Lasserre [20] extract optimizers and detect global optimality, and Nie, Demmel, and Sturmfels [21] give SOS exactness over the gradient ideal. Exact rational certificates recovered from a numerical SDP are due to Peyrl and Parrilo [8], Kaltofen et al. [9], and Magron and Safey El Din [10]; closest to our rate statement, Davis and Papp [11] construct exact rational certificates from numerical dual certificates and prove a convergence rate. Rigorous, though not exact, bracketing of the relaxation value by verified SDP is due to Jansson [12] and Roux et al. [13], and an exact SDP duality theory to Ramana [22]. A nonconvex QCQP is the degree-two instance of this machinery; its duality, exactness, and hidden convexity are classical [6, 23, 7, 14, 15, 24], as is the trust-region hard case [16, 25].

Against this background our contribution is narrow. We instantiate the exact rational recovery of Peyrl and Parrilo and of Davis and Papp on the S-procedure dual matrix $M(\lambda, t)$ of a QCQP; we record the rounding rate of the resulting certified lower bound, second-order in the nonsingular case (Theorem 2) and first-order in the trust-region hard case (Theorem 3). This is the standard second-order-versus-first-order behavior of rounding at an interior as against a boundary maximizer, stated for the QCQP dual; we do not claim it as new. The weak-duality bracket and its closure are likewise standard. The contribution is the integration: a tolerance-free, exact-rational instantiation of this machinery, wired into a guidance and control verification stack (Section 4.5) and re-checked automatically. Barrier and control-barrier certificates [3, 4], credible autocoding [5], and the verification of convex-optimization solvers for model-predictive control [26] are complementary in verifying control software: the last verifies the solver algorithm, whereas we verify its output. Lossless convexification [18] and successive convexification [1] address the feasibility and tightness of convex reformulations; the optimality-gap bounds here certify the global suboptimality of the resulting trajectory (Section 4.5).

7 Limitations and open problems

The lower-bound soundness of Theorems 1 and 4 and the nonsingular recovery of Theorem 2 are complete. An exact rational recovery in the singular case (Theorem 3) and an exact higher-level moment certificate for the multi-constraint gap (Theorem 4) remain open. The $LDL^{\top}$ PSD test is $O(n^3)$; on the per-primitive problems these checkers target the rational entries stay small, and they grow with problem size under exact elimination, which is why the certificates are applied per primitive rather than to arbitrary sizes.

8 Conclusion

We have described a method that verifies the output of untrusted guidance and control computation in exact rational arithmetic, and applied it to barrier safety, KKT optimality, Lyapunov stability, and sum-of-squares set containment. Its technical content specializes exact rational SDP-certificate recovery to the S-procedure dual of a nonconvex QCQP, and characterizes the rounding rate of the certified lower bound, which is second-order in the nonsingular case and first-order in the trust-region hard case, with soundness retained in both, and a certified exact duality gap when the relaxation is not tight. The constructions are realized in an open reference implementation.

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